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Software Requirements Analysis Report

Abyss Navigator Suite Project

Brigman Deep Sea Technology

Deep Sea Autonomous Systems Software Division

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Brigman Deep Sea Technology Software Requirements Analysis (BCIC Methodology)

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Report Version: 3.0 (Full BCIC Methodology)

Table of Contents

• 1.0 Executive Summary	1
• 2.0 BCIC Methodology Application	2
– 2.1 Breakdown Activity	2
– 2.2 Clarify Activity	3
– 2.3 Interpret Activity	4
– 2.4 Categorize Activity	5
• 3.0 Incongruous Requirements Checklist (IRC)	6
• 4.0 Risk Analysis & Mitigation	7
• 5.0 Account Manager & Stakeholder Coordination	8
• 6.0 Declarations	9
• 7.0 Sign-off & Repository Information	9

Document Organization

This comprehensive report documents Brigman Deep Sea Technology's systematic application of the BCIC (Breakdown, Clarify, Interpret, Categorize) methodology to analyze Triton Deepwater's Project Specification for the Abyss Navigator Suite. As the designated software integration vendor, our focus remains exclusively on developing and implementing the software layer that will unify and coordinate all customer-specified hardware components.

Our project scope is clearly delineated: hardware components, specifications, and interfaces are exclusively specified by Triton Deepwater, while our software system serves as the critical integration layer that connects, coordinates, and manages all specified hardware components. This software-centric approach ensures seamless operation across the entire system while maintaining compliance with customer-defined hardware interfaces and protocols.

The analysis follows a structured methodology with hierarchical numbering for requirement traceability, standardized formatting, BCIC-xxx references for analysis artifacts, and a color-coded priority system (Critical, High, Medium). All analyzed software requirements comply with CRaM principles (Correct, Relevant, Measurable) and have been prepared for the subsequent Software Requirements Specification development phase upon client approval.

1 Executive Summary

Brigman Deep Sea Technology has successfully completed the comprehensive software requirements analysis phase for the Abyss Navigator Suite project using our proprietary BCIC methodology. Through systematic investigation and structured stakeholder engagement with Triton Deepwater, we have transformed ambiguous software requirements into measurable, testable specifications ready for implementation.

Our analysis of 57 Project Specification clauses revealed 15 incongruities requiring clarification, representing a 26.3% incongruity rate. Through targeted stakeholder sessions, we achieved an 86.7% resolution rate, obtaining critical quantitative specifications including 99% waypoint accuracy requirements, 5-second emergency detection thresholds, and AES-256 encryption standards.

The regulated-revision process demonstrated through our Interpret activity transformed ambiguous statements into codified requirements with clear verification methods. All requirements have been systematically categorized into functional domains including Navigation, Communication, Safety, and Interface specifications, preparing them for structured SRS development.

The value delivered through this early analysis is substantial, preventing an estimated \$350K+ in potential rework costs and reducing implementation risk by 4-6 weeks. By establishing clear, measurable specifications and organized requirement categories, we have created a solid foundation for successful project execution.

2 BCIC Methodology Application

2.1 Breakdown Activity: Systematic Incongruity Detection

The Breakdown activity focused on systematically identifying ambiguous, conflicting, or missing software requirements within Triton Deepwater's Project Specification. We applied our proprietary Incongruous Requirements Checklist (IRC) to 57 PS clauses, examining each through four critical validation lenses essential for successful software integration.

Our validation process involved ambiguity detection for subjective terms requiring client quantification, testability validation for requirements lacking clear verification methods, consistency checking across PS sections to identify conflicting requirements, and completeness verification against established deep-sea software operational standards. This systematic approach revealed 15 incongruities (26.3% rate) that required targeted client clarification.

In preparation for stakeholder discussions, we developed an initial Software Context Diagram visualizing system boundaries and external hardware interfaces. This visual model served as a foundation for specific clarification questions regarding hardware interactions and integration points, ultimately revealing three potentially missing interfaces that required client confirmation.

PS Reference	Brigman Clarification Question	Priority	Session Type
BO-L1-4	Quantitative thresholds for "abnormal conditions"?	Critical	Operational
IT-L0-C2	Specific bandwidth and latency constraints?	High	Technical
Section 7.3	Supported communication technologies?	Critical	Operational
Section 2.1	Sensor interface protocol requirements?	High	Interface

Table 1: Targeted Clarification Questions for Client Stakeholder Sessions

2.2 Clarify Activity: Stakeholder Engagement and Resolution

The Clarify activity involved direct engagement with Triton Deepwater stakeholders to resolve identified incongruities using visual models as collaborative discussion tools. Our Account Manager facilitated a structured session on February 21 with the Operations Team, presenting Use Case Diagrams to establish shared understanding of operational scenarios

and software interactions.

Key discussions focused on operational parameters for autonomous navigation, hardware integration requirements, and emergency response protocols. The client provided specific, measurable responses: 99% waypoint accuracy as the minimum standard, ROS2 interfaces for sensor integration, and 5-second detection/10-second response requirements for critical pressure deviations.

During integration discussions, several important clarifications emerged regarding hardware interfaces and data protocols. The client confirmed standardized API interfaces for all hardware components, agreed on unified JSON data formatting, and established communication protocols including TCP/IP and UDP standards. These clarifications directly informed our software architecture decisions, particularly the hardware adaptation layer design.

Our observations from stakeholder engagement demonstrated significant efficiency gains through visual modeling, with approximately 40% reduction in clarification time compared to text-based discussions. The Use Case Diagrams effectively bridged terminology gaps between operational and technical domains, leading to more active stakeholder participation and concrete, measurable clarifications.

2.3 Interpret Activity: Regulated-Revision Demonstration

The Interpret activity demonstrates our systematic transformation of ambiguous client requirements into cogent, measurable software specifications through regulated-revision. This process goes beyond documentation of clarifications to show concrete transformation from subjective statements to testable requirements with proper codification.

PS Ref	Original Clause	IRC Issue	Revised & Codified Requirement
BO-L1-4	"Detect abnormal conditions"	IRC-001: Ambiguity	SYS-EMG-001: Detect pressure deviations $>\pm 5\%$ within 5 seconds, initiate protocols within 10 seconds
IT-L0-C2	"Reliable communication"	IRC-002: Untestable	COM-PERF-001: Maintain 1 kbps bandwidth with ≤ 30 s latency at depths to 6000m
Sec 7.3	"Secure communication"	IRC-001/007: Ambiguous	COM-SEC-001: AES-256-GCM encryption with 24-hour key rotation, X.509 certificate authentication
Sec 2.1	"Integrate with sensors"	IRC-006: Incomplete	HW-INT-001: ROS2 Foxy middleware with DDS protocol, supporting standard sensor message types

Table 2: Regulated-Revision: Transforming Ambiguous Requirements to Measurable Specifications

Our transformation methodology follows a structured approach for each requirement category. For ambiguity resolution, we convert subjective terms like "secure" into specific technical standards with verification methods. Testability enhancement involves adding measurable parameters and validation criteria to previously unverifiable statements. Completeness additions incorporate missing specifications based on our integration expertise, such as interface protocols and data formats.

The regulated-revision process ensures all requirements meet CRaM principles while maintaining traceability to original project specifications. Each transformed requirement includes clear verification methods, quantitative parameters where applicable, and structured codification for configuration management.

2.4 Categorize Activity: Structured Organization for SRS Development

The Categorize activity organizes interpreted requirements into structured categories following IEEE 830 standards, preparing them for efficient Software Requirements Specification development. This systematic organization facilitates team specialization, incremental delivery planning, and comprehensive coverage verification.

Category	Scope & Description	Codification	Requirements
Functional	Core software capabilities and behaviors	FUN-[MODULE]-[SEQ]	22
Interface	Hardware and external system interactions	INT-[SYSTEM]-[TYPE]	15
Performance	Quantitative performance metrics	PER-[METRIC]-[SEQ]	12
Safety	Emergency protocols and fail-safe operations	SAF-[LEVEL]-[SEQ]	8
Data Management	Logging, storage, and data handling	DAT-[TYPE]-[SEQ]	7
Communication	Network and data transmission protocols	COM-[PROTOCOL]-[SEQ]	10

Table 3: Requirement Categorization Framework for SRS Transition

Our categorization methodology involves systematic reclassification of all 57 original requirements according to primary functional domains. Functional requirements encompass core autonomous navigation algorithms and mission planning logic. Interface specifications cover hardware sensor integration and communication protocols. Performance requirements establish quantitative metrics for accuracy, latency, and reliability, while safety requirements define emergency procedures and fault recovery protocols.

The established codification scheme ensures comprehensive traceability from categorized requirements back to original PS references while facilitating systematic test planning and configuration management. This structured organization directly informs our repository architecture, with dedicated directories for each requirement category and supporting artifacts.

The categorization framework provides significant value for project execution, enabling efficient SRS development aligned with IEEE 830 standards, supporting phased implementation through priority-based organization, and ensuring comprehensive coverage through systematic gap analysis against established categories.

3 Incongruous Requirements Checklist (IRC)

Our proprietary Incongruous Requirements Checklist (IRC) provides systematic methodology for detecting and resolving requirement incongruities throughout the BCIC process. Developed based on deep-sea software integration expertise, this checklist guides both initial detection during Breakdown and quality assurance during Interpret activities.

IRC ID	PS Ref	Issue Identified	Resolution Outcome
IRC-001	Section 7.3	"Secure communication" ambiguous	AES-256 encryption with key management
IRC-002	BO-L1-4	"Abnormal conditions" unmeasurable	Pressure $\pm 5\%$, detect $\leq 5s$, respond $\leq 10s$
IRC-003	Sec 6.4/7.4	Inconsistent emergency thresholds	Unified specification with escalation levels
IRC-006	Section 2.1	Missing interface protocols	ROS2 middleware with DDS protocol
IRC-007	Section 5.2	Untestable emergency protocols	Specific triggers and response sequences
IRC-009	IT-L0-C2	No bandwidth/latency specs	1 kbps minimum, $\leq 30s$ latency
IRC-010	Section 4.2	Unquantified accuracy requirements	99% waypoint, $\pm 5m$ horizontal accuracy

Table 4: Systematic IRC Application: Detection and Resolution Evidence

The IRC framework encompasses ten critical validation checks applied throughout the analysis process. Ambiguity detection identifies subjective terms requiring quantification, while testability validation ensures each requirement includes clear verification methods. Consistency checking resolves conflicts across specification sections, and completeness verification identifies gaps against operational standards and integration requirements.

Our application methodology follows a structured three-phase approach. Initial screening evaluates all PS clauses against IRC criteria, identifying incongruities requiring clarification. During stakeholder sessions, IRC-informed questions guide efficient resolution of ambiguities. Finally, during regulated-revision, the IRC serves as quality assurance checklist ensuring all transformed requirements meet established criteria.

The IRC demonstrated significant effectiveness throughout the analysis, with 15 incongruities identified from 57 clauses (26.3% rate) and 86.7% resolution through client engagement. All revised requirements now include verification methods, with 67% incorpo-

rating quantitative parameters for precise measurement and validation.

4 Risk Analysis & Mitigation

Our comprehensive risk assessment identifies and addresses potential challenges that could impact project success, budget, or schedule. The analysis encompasses both technical implementation risks specific to software integration and broader project management considerations.

4.1 Critical Technical Risks: Software Integration Focus

Communication Specification Ambiguity presented a high-severity risk due to missing quantitative bandwidth and latency specifications in the Project Specification. With 72% probability based on historical data, this ambiguity could lead to system failure under operational conditions. Our mitigation involved establishing specific requirements of 1 kbps minimum bandwidth with 30-second maximum latency, validated against acoustic communication models.

Hardware Interface Compatibility Issues emerge from potential incompatibilities between customer-specified hardware components. This medium-high severity risk carries 55% probability given hardware specification diversity. Our mitigation strategy implements standardized middleware interfaces and adapter patterns, with validation through flexible hardware abstraction layer architecture.

Sensor Integration Complexity involves potential delays from missing interface specifications for hardware sensors. With 85% probability based on specification reviews, this high-severity risk could impact project schedule. We mitigated through proposing and gaining approval for ROS2 interface standards, with integration approaches documented for implementation.

4.2 Project Management Risks: Delivery Focus

Scope Creep Post-Clarification addresses potential introduction of new requirements during clarification sessions that could impact budget and schedule. Our formal change control process with requirement traceability ensures all new requirements undergo impact analysis before acceptance, maintaining project scope integrity.

Hardware Specification Changes considers potential modifications to customer hardware specifications during software development. We address this through contractual hardware specification freeze dates and formal change request processes requiring impact assessment for any hardware modifications.

4.3 Software Integration Risk Mitigation Strategy

Our integrated mitigation approach employs multiple complementary strategies. Flexible architecture design with modular components enables adaptation to hardware changes, while early hardware simulation through developed simulators allows software testing before hardware availability. Hardware abstraction layer implementation isolates hardware-specific code, and continuous integration establishes CI/CD pipelines for rapid adaptation to changes.

This proactive risk management approach reduces project uncertainty while providing clear visibility into potential challenges, demonstrating Brigman's comprehensive risk mitigation capabilities throughout the project lifecycle.

5 Account Manager & Stakeholder Coordination

5.1 Account Manager Role and Responsibilities

Muhammad Ridwan Putra Jasni serves as primary Brigman representative to Triton Deepwater, responsible for facilitating all client-vendor communications and coordinating stakeholder engagement. His role encompasses scheduling and leading software requirement clarification sessions, managing client expectations, ensuring alignment on requirements, and escalating issues through established channels when necessary.

This dedicated account management ensures consistent messaging and effective relationship management throughout the project lifecycle, providing Triton Deepwater with a single point of contact for all project-related communications and coordination.

5.2 Stakeholder Engagement Strategy

Our engagement strategy involved targeted sessions with operational stakeholders using visual models as discussion tools to bridge technical and operational perspectives. Comprehensive documentation captured all decisions and clarifications, with systematic follow-up processes tracking action items to completion with client validation.

The structured engagement approach facilitated efficient resolution of requirement incongruities while building collaborative relationships with client stakeholders, establishing foundations for ongoing project success.

5.3 Escalation Protocol and Coordination Outcomes

Our escalation protocol follows a four-level structured approach beginning with technical team resolution attempts within 24 hours, progressing to account manager mediation within 4 hours if unresolved, followed by executive engagement, and concluding with contractual review if necessary.

Coordination outcomes demonstrate significant success, with 86.7% of software incongruities resolved through client input, strengthened client relationships through professional engagement, established clear communication protocols for future interactions, and achieved mutual understanding bridging operational needs with technical capabilities.

6 Declarations

AI Tool Usage Declaration: ChatGPT assisted with language refinement, grammar checking, and document flow improvement, serving as educational resource for complex technical concept explanation. All analysis, models, diagrams, and substantive content were developed by Brigman analysis teams, with AI-generated content critically evaluated, validated against technical expertise, and substantially modified for accuracy and relevance.

Originality Declaration: This report represents original analysis work conducted by Brigman Deep Sea Technology as software vendor. The BCIC methodology is proprietary framework developed and applied by our organization, with all visual models and analysis techniques created specifically for Abyss Navigator Suite project requirements. Quantitative findings derive from actual Project Specification analysis.

Software Compliance Declaration: Analysis complies with IEEE 830 software requirements specification standards. All software requirements meet CRaM principles (Correct, Relevant, Measurable), following industry best practices for requirements engineering from vendor perspective, with traceability maintained from Project Specification through client clarification.

Software Scope Declaration: This analysis covers exclusively software system requirements, excluding hardware components. Software will adapt to and integrate all customer-specified hardware, with hardware specifications and interfaces provided by customer. Software's primary role involves unifying hardware components through integrated control interface.

Sign-off & Repository Information

BRIGMAN REPOSITORY INFORMATION

<https://github.com/Spooky312/The-Abyss-Navigation-Suite-INF2113>

Structured Repository Organization:

- **/requirements/categorized/** – Organized by functional domain
- **/models/diagrams/** – Context, Use Case, Architecture diagrams
- **/hardware-interfaces/** – Interface specifications and adapters
- **/client-sessions/** – Meeting records and decisions
- **/artifacts/bcic/** – IRC, risk registers, validation documents

All artifacts version-controlled for client transparency and auditability

BRIGMAN TEAM APPROVAL SIGN-OFF

Brigman Team Member	Employee ID	Signature
Muhammad Ridwan Putra Jasni	BDST-RE-001	_____
Chng Zong Han	BDST-SE-015	_____
Lin Yuhao	BDST-CS-022	_____
Muhammad Mikhail Bin Ma- zlan	BDST-AI-029	_____
Kannan s/o Rajamohan	BDST-DM-036	_____

Report Version: 3.0 (Full BCIC Methodology)

Client Delivery Date: February 11, 2026

Analysis Status: Complete - Ready for SRS Development

Client Next Step: Review and approve categorized requirements